Inverse kinematics of a subsea constrained manipulator based on FABRIK-R

*Phillipe Santos, Matheus Santos, Satja Sivčev, Edin Omerdic, Gerard Dooly, Daniel Toal
Centre for Robotics & Intelligent Systems
University of Limerick
Limerick, Ireland
phillipe.santos@ee.ufcg.edu.br

***Abstract*—Although underwater manipulators are considered the most suitable tools for executing many subsea intervention operations, they are an example of robots not designed for autonomous tasks. The subsea manipulator Schilling Titan 2, for instance, is composed of a sequence of constrained joints that makes impossible finding a closed-form analytical solution for its inverse kinematics. For this reason, numerical methods have normally been used. Nevertheless, these methods typically result in high computational load and may need many iterations to find a solution. FABRIK is an inverse kinematics approach which has as main advantages a fast convergence and a low computational cost, avoiding matrix inversion and being robust to singularities. However, the constraints imposed by the joints of the Titan 2 manipulator also makes impossible solving its inverse kinematics using FABRIK. A recent extension of FABRIK, named FABRIK-R, presents a solution for this problem. Therefore, this paper proposes to overcome the mentioned issues by solving the inverse kinematics of a high constrained subsea manipulator based on FABRIK-R and presents a comparison between this solution and a classical approach based on a second-order pseudo-inverse Jacobian. The advantages of the proposed approach are demonstrated in simulation experiments, and its feasibility is demonstrated in real experiments in manipulation tasks performed in dry lab conditions.**

***Index Terms*—Manipulator, FABRIK, Inverse Kinematics, Robot, Constraints.**

I. Introduction

Manipulator robots have been used in industry applications ranging from pick-and-place operations to assembling complex electronic devices. Nevertheless, the application of manipulator robots is not limited to industry, but it has reached disparate areas such as cooking, rehabilitation, education and even surgery [1]. The incorporation of robots in other areas added new requirements for manipulators, such as the demand for more flexibility and intelligence to perform different tasks autonomously [2].

The autonomy of movement to position the end-effector in a specific pose is given by the inverse kinematics, which is defined in [3] as the problem of determining an appropriate set of joint configurations so that the end-effector moves to desired positions smoothly, quickly and accurately.

However, some robots are not designed for autonomous robotic tasks and the way their serial chain is designed bring issue for the mathematical modelling process, including solving inverse kinematics. According to [4], although the underwater manipulators are considered the most suitable tools for executing subsea intervention operations, they are an example of robots not designed for autonomous tasks. The Schilling Titan 2 manipulator, for instance, does not have a spherical wrist, and therefore finding a closed-form analytical solution is impossible. For that reason, the authors solved the inverse kinematics problem for the Titan 2 manipulator using numerical methods, since they are model independent and applicable to any robotic manipulator.

Many different numerical methods have been applied to manipulators. The best known are the Jacobian methods that linearly model end-effector movement in relation to changes that occur in the joint angles. Within this group, several approaches have been developed to calculate or approximate the inverse of the Jacobian, such as the transposed Jacobian, Damped Least Squares (DLS), Singular Value Decomposition (SVD), and various extensions [5]. The algorithm employed in [6] is the closed-loop second-order inverse kinematics algorithm with pseudo-inverse Jacobian [7]. The solutions presented based on the Jacobian matrix produce smooth positions, however, most of these approaches suffer from high computational cost, complex matrix calculations and singularity problems.

Other numerical approaches are based on Newton's methods for problem solving. The Broyden, Powell, and the BFGS methods can be highlighted [8] [9]. Newtonian methods are presented as a minimization problem, so they return smooth motion without erratic discontinuities. In addition, they also have the advantage of not suffering from uniqueness problems, as occurs in Jacobian methods. However, these approaches are complex, difficult to implement, and have a high computational cost per iteration [3].

Numerical methods typically have a high computational cost and may need many iterations to find a solution, consequently, the time needed to solve the problem is also high. For some applications, such as the one presented in [6], in which the manipulator has to track a moving target with a camera mounted on the end-effector, taking too long to process the manipulator pose can lead to the target object leaving the field of view of the camera, and the tracking failing.

Heuristic-based methods are the simplest and fastest numerical approaches. The main methods in this category are Cyclic Coordinate Descent (CCD), Triangulation, and FABRIK [10]. CCD is introduced in [11]. This method attempts to minimize position and orientation errors by iteratively transforming

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joint variables, one at a time. However, CCD often biases in describing unrealistic movements for robot manipulators.

The triangulation method, presented in [12], corresponds to a non-recursive technique to solve the problem based on the cosine law. However, many constraints are not addressed, such as joint limits and end-effector orientation control.

The FABRIK solution, proposed by [13], consists of an iterative method that treats each joint value as a point in the Cartesian plane and thus searches for subsequent joints as points belonging to lines, representing the manipulator links. This approach has the main advantages of a high convergence percentage and low computational cost, since it does not make use of matrix compositions and inversions.

In addition, an extension of FABRIK, presented in [14], suggests solutions to integrate constraints to the original algorithm, in order to deal with different kind of joint constraints.

Due to its advantages, FABRIK can be considered as a powerful tool to solve the problems regarding the processing time to calculate the inverse kinematics of robotics manipulators.

However, the method proposed in [14] consists of creating a constraint plane which cannot be easily defined for kinematic chains with only 1-DOF joints, making its application unfeasible for highly constrained robotic manipulators, such as the Titan 2. In [15], we proposed an extension of FABRIK to deal with this problem. The extension, called FABRIK-R, was tested with a real robot of 7 DOF.

Therefore, in this paper we propose to solve the inverse kinematics of a highly constrained subsea manipulator based on FABRIK-R and present a comparison between this solution and a classical approach based on the second order of the pseudo inverse Jacobian. The comparison aims to highlight the advantages of applying this simpler and faster method for real applications on underwater manipulators. The approach has been successfully tested in simulation and in real experiments in dry lab conditions.

The paper is structured as follows. In section 2, we present the visual servoing control developed in [6], highlighting the bottlenecks imposed by the use of a numerical approach. In section 3, we introduce the original algorithm FABRIK. Then, in section 4, we present our adapted FABRIK-R for the use with the Titan 2 manipulator. The results with the comparison between our method and the numerical approach are presented in section 5 and conclusions in section 6.

II. Numerical approach limitations in a PBVS system

Position-based visual servoing (PBVS) is the use of machine vision data to estimate target pose in the Cartesian space in order to control the motion of a robot [16]. Based on the target position estimation and the arm pose obtained from the joint position sensors, the target pose is computed at the robot base frame by homogeneous transformations. Then, the inverse kinematics algorithm is performed in order to calculate the new joint positions to be forwarded to the existing low level motion controller.

Sivčev et. al. [6] used an eye-in-hand configuration, with a camera mounted on the wrist joint of the Titan 2 manipulator and the target position was estimated based on fiducial markers. The authors performed the inverse kinematics by means of the closed loop second-order inverse kinematics algorithm with pseudo inverse Jacobian proposed in [17]. This approach was successfully tested for some underwater intervention tasks such as the one presented in Fig. 1, in which the Titan 2 manipulator was used for a valve turning test.

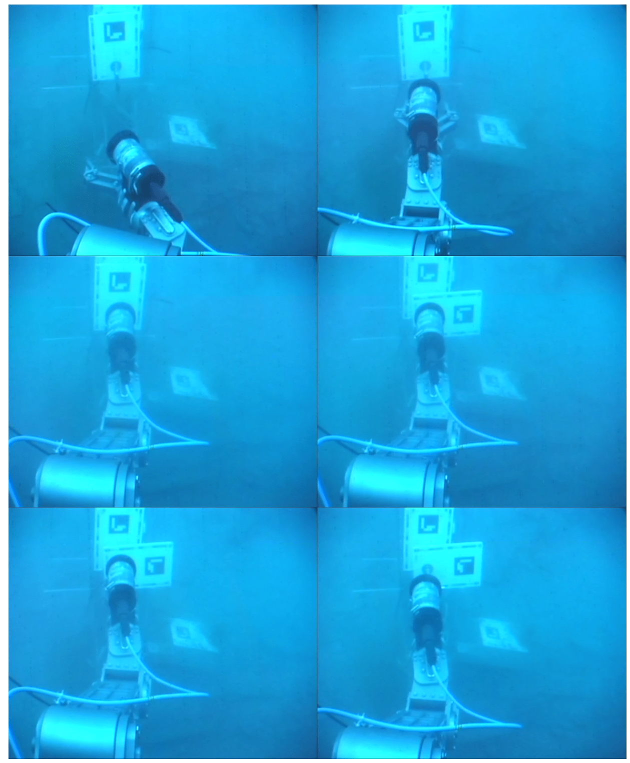

Fig. 1. Automatic valve turning with Schilling Titan 2 [6]

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Although the experiments were successful, Sivčev et. al. mention problems with the amount of time taken by the inverse kinematics approach which directly affects the frequency of target pose estimation.

Approaches based on Pseudo-inverse Jacobian normally need a high number of iterations to find a solution. This high number of iterations can inevitably result in the inverse kinematics algorithm execution taking too much time and the object can leave the camera field of view before the next image is taken for pose estimation. This causes the opening of visual feedback because of the lack of visual measurements and eventually to servoing failure [4]. An option to overcome this might consider to reducing the number of iterations to keep the object in the field of view of the camera during the task execution. However, having insufficient number of iterations will, due to the higher mismatch of the inverse kinematics output and the given desired value, increase the total number of outer iterations which will increase the total time required for the task to be executed.

In [4] to overcome the problem intermediary, points are generated in the cartesian space, between the initial and final pose. This is done in steps covering a certain amount of the path towards the target. However, this approach inevitably increases the total time required for the task to be executed.

Therefore, it is clear that a faster inverse kinematics approach would improve the performance of the PVBS algorithm. Thus, we present a new approach based on the algorithm FABRIK-R to overcome this convergence time bottleneck.

III. Forward and Backward Reaching Inverse Kinematics (FABRIK)

FABRIK is an algorithm that works by controlling the joint angles to minimize the error between the target point and the end-effector. The control is carried out by adjusting the joints one at a time [13], firstly in the end-effector to base direction, and subsequently in the base to end-effector direction.

Fig. 2 presents an illustration of the method considering a three link planar manipulator for a better understanding. In this example, p_1, p_2, p_3 and p_4 represent the joints of a planar manipulator d_1, d_2, d_3 and d_4 represent the links, and the letter t indicates the target point.

The first step in the algorithm consists of checking whether the target point is reachable, by summing up all the link lengths. If reachable, the algorithm is performed in two stages. In the first stage, the algorithm estimates each joint position successively, starting from the end-effector going to the manipulator base. Considering the example in Fig. 2, where p_1 is the base and p_4 the end-effector, the first step of this stage is to assign the end-effector to a new position p_4', which corresponds to the target point t (Fig. 2b). The next step is to draw a line between p_4' and p_3. The new position of joint p_3, lies on this line, at a distance equal to the length of the link d3, (Fig. 2c). Similarly, the new position of joint p_2, is given on a line drawn between p_3' and p_2, at a d2 link distance (Fig. 2d). Finally, the process is repeated to find a new position for the base p_1' (Fig. 2d).

As shown in Fig. 2d, at the end of the first stage of the algorithm, the manipulator base can be out of its initial position. Thus, a second stage of the algorithm is necessary for the base to return to its original position. This second stage is called Backward Reaching and the same procedure to that in the Forward Reaching is repeated, but this time starting from the base and moving to the end-effector. The first step is presented in Fig. 2e, by returning the joint p_1' to the base initial position, being labeled now as p_1''. Then, a line is drawn from p_1'' to p_2' and the new position p_2'' is defined based on the link length d_1.

After a complete iteration (Forward and Backward), the end-effector will always be closer to the target [13]. Thus, the procedure must be repeated iteratively, as many times as necessary, until the end-effector reaches the target within a user given precision.

Although FABRIK is presented with a planar manipulator with 4 degrees of freedom, the algorithm works the same way in a three-dimensional space with manipulators with any

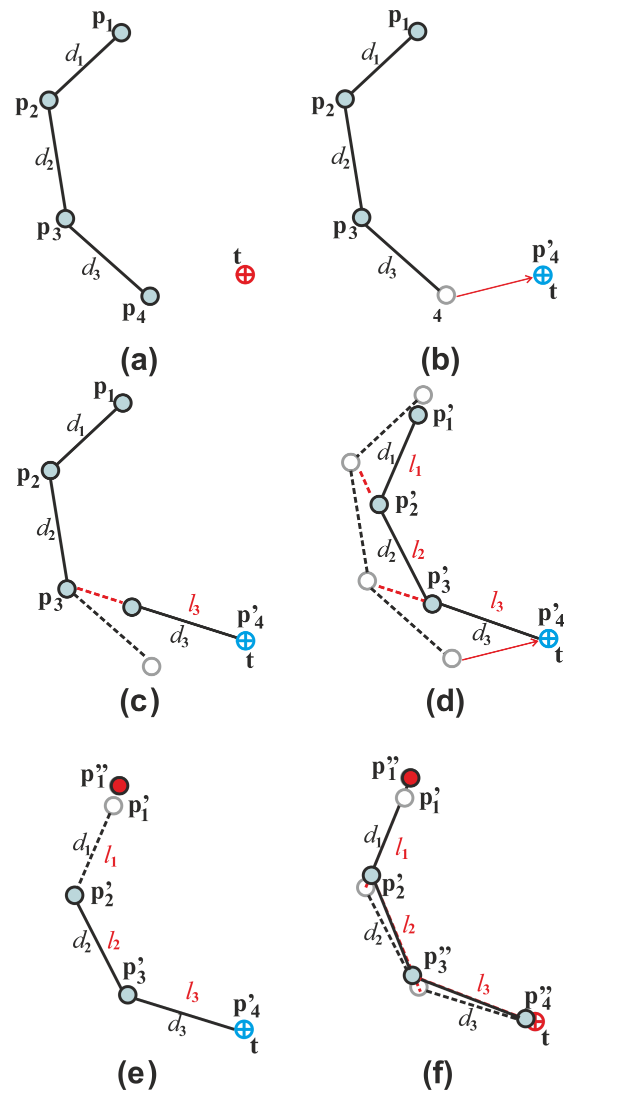

Fig. 2. An example of Forward Reaching (FR) and Backward Reaching (BR) phases of FABRIK for a 3-DOF robot [13]. a) Initial and target position. b) FR: Moving end-effector to target position. c) FR: New position of joint 3. d) FR: New position of all joints and end-effector. e) BR: Moving base to initial position. f) BR: New position of all joints.

number of joints. FABRIK has also been extended to consider different types of joint models and their constraints [10].

Other authors have also presented approaches based on FABRIK. In [18], for example, a variation of FABRIK is used to solve the problem of inverse kinematics from an energy transfer perspective. In [19], the algorithm is adjusted to work with joints that have more than two segments. In [20], a data driven prior to warm start version of FABRIK is used to produce more natural human poses. In [21], a collision-free motion planning approach was proposed based on FABRIK. In [22] a control system is developed combining FABRIK and a PID controller for a simple and fast tracking approach with the end-effector.

Although many different works have been published based on FABRIK variations, none of the approaches reviewed have been applied to a highly constrained arm, except for the FABRIK-R [15].Therefore, we present the inverse kinematics of the highly constrained subsea manipulator Titan 2 based on FABRIK-R in the next section.

IV. Adapted FABRIK for the subsea manipulator Titan 2

In this section, we firstly introduce the approach used in [13] to contemplate the movement constraint imposed by a hinge joint on the FABRIK algorithm. Then, we highlight the lack of important information for the application of this approach to manipulators and propose a solution for it.

Aristidou et. al. present their approach using a two link robot in 3D space, as presented in Fig. 3. In this example, the chain is composed of only three points: the hinge joint (p_2), in which the constraint is applied, the root (p_1) and the target (p_3). The method works by defining two planes. One that represents the allowed motion (Φ_1) defined by the hinge joint $p2$ (1 DoF), and the other (Φ_2), which is oriented, defined by the root and the target. Then, the basic idea is to project the manipulator to the plane Φ_2, reorienting the hinge joint p_2 so that the plane of movement is aligned to the target. However, in their paper, the authors do not specify clearly how to apply this method to a body with more than 3 joints or how to define which joint is the root and the target for this case. The lack of definition allows different interpretations on applying the described approach for a given p_i at the forward reaching stage, such as:

1) Set root as the base of the manipulator and target as the end-effector.
2) Set root as p_{i-1} and target as p_{i+1}.
3) When in a hinge chain, apply the method joint after joint, i.e, sequentially constrained, p_i, p_{i-1}, p_{i-2} until the end of the chain.
4) When in a hinge chain, set p_i and p_{i-1} at once, i.e, sequentially constrained, p_i, p_{i-2}, p_{i-4} until the end of the chain.

The definition of Φ_i plane may also introduce a problem, as it neglects possible movement restrictions in the joints connected to the hinge. Thus, the method can be functional only if the root and the target have total freedom of movement. This restriction is possibly not considered in the algorithm because FABRIK was originally developed for computer graphics applications. However, when considering the use of FABRIK on Titan 2, the application of the method as presented in [10] is impracticable, since the kinematic chain is composed by joints of only 1DOF on the Titan 2.

The Titan 2 robot is composed of a pivot joint at the base, four hinge joints in the kinematic chain and another pivot joint at the end, as shown in Fig. 4 with joints rotating around the z_i axes. These hinge joints are configured in a sequence of three joints in the same actuation plane, p_2, p_3 and p_4 and another joint p_5 in a plane orthogonal to the actuation plane of other 3 hinge joints.

Therefore, some modifications are needed in the method described in Fig. 3 to allow the application of FABRIK on the

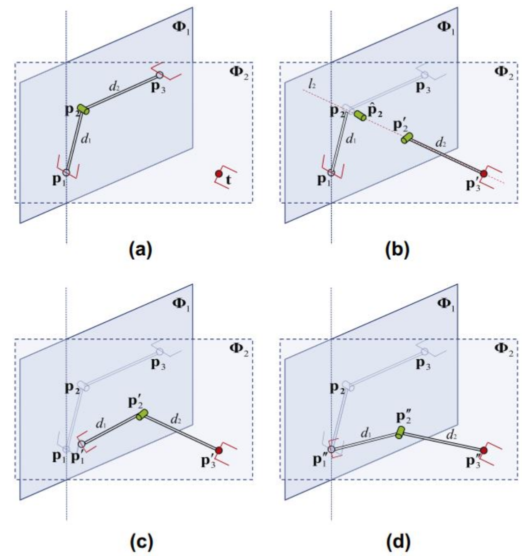

Fig. 3. Applying hinge joint constraints for FABRIK algorithm. (a) Initial configuration and target. Φ_1 represents the plane where p_2 joint movement is allowed. Φ_2 is set by the root and target, which is oriented. (b) Reallocate and reorient the p_3' joint to the target t. Then, project p_2 onto the plane Φ_2, generating $\widehat{p}_2$ and find p_2' on line l_2 which passes from the joint p_3' and point $\widehat{p}_2$ and has distance d_2 of p_3'. Reorient the new joint according with Φ_2. (c) Move and reorient p_1 to p_1' which is on the line l_1, between the joint p_2' and p_1 and has distance d_1 from p_2'. (d) Now it is a 2D problem, since all joints are in the plane. [10]

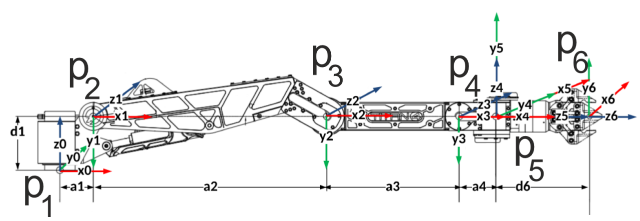

Fig. 4. Kinematic model of robotic manipulator with Denavit-Hartenberg convention coordinate frame assignment [6].

Titan 2. These modifications are made based on the FABRIK-R method [15] and we illustrate its application Fig 5. The first step of FABRIK-R's algorithm is to set the end-effector to the target, at the position p_6', and set the new position of joint 5, p_5', that lies on the line between p_6' and the joint p_5, at a distance from p_6' corresponding to the link d_5 length.

The problem now is how to define the new position of p_4, considering that its position must obey not only the constraint imposed by the joint p_5', which is represented by Φ_5, but also the constraints imposed by the previous joints p_3, p_2 and p_1. Since p_4, p_3 and p_2 have the same actuation plan, it is not possible to define p_4' by creating a plane based on p_3, as proposed in [13]. Thus, we create a new plane Ω using the plane Φ_5 passing through point p_5' and the joint p_1, which is the only joint able to change the actuation plane of p_4.

After the definition of the plane Ω, the other points of the kinematics chain follow steps similar to Fig. 3 (b) and (c). The joint p_4 is projected onto the plane Ω, generating $\widehat{p}_4$, and then p'_4 is found on the line l_4 which passes from the joint p'_5 and the point $\widehat{p}_4$ and has distance d_4 from p'_5. The joint p'_4 is reoriented according to Ω. The remaining joints are set by the same process in a way that at the end of the forward reaching (Fig. 5b), the joints p'_4, p'_3 and p'_2 have the same actuation plane $\Phi_{2,3,4}$, which is the same plane Ω created to impose the constraints in the algorithm. However, this plane $\Phi_{2,3,4}$ might be inclined in a direction impossible to be positioned by p_1. Thus, the Backward Reaching starts by moving the point p_1, which describes the base of the robot, to its initial origin since it has moved with the forward reaching process, and correcting the actuation plane $\Phi_{2,3,4}$ to be perpendicular to the plane xy. As the first joint is a pivot joint, the plane $\Phi_{2,3,4}$ can be freely rotated around the z-axis to contain the points p'_5, which is the joint with a different actuation plane. Once the plane is defined, as presented in Fig. 5c, p_2, p_3 and p_4 are projected on it and the new positions for joints p_2 to p_6 are allocated according the the FABRIK algorithm explained in section 3.

The adaptations made by the FABRIK-R allow the method to converge to a solution considering movement constraints. The method has been implemented and tested by simulation and on the real robot Schilling Titan 2, and the results are presented in next section.

V. Results

This section shows the application of FABRIK-R for the underwater manipulator Schilling Titan 2, presented in Fig. 6, and compares it with the second order Pseudo-inverse Jacobian approach proposed by [6]. The robotic arm is mathematically modelled as an open kinematic chain of seven rigid links connected by six joints using the DH convention presented in table I. The coordinate frames (O_{x0y0z0} to O_{x6y6z6}) assigned to the manipulator joints are presented in Fig. 4.

Firstly, we compare the performance of the algorithms based on a set of one thousand target points randomly distributed in the reachable space. In these tests, the positioning error tolerance is set to 1 cm. Table II presents the average runtimes of both approaches, as well as the number of iterations needed

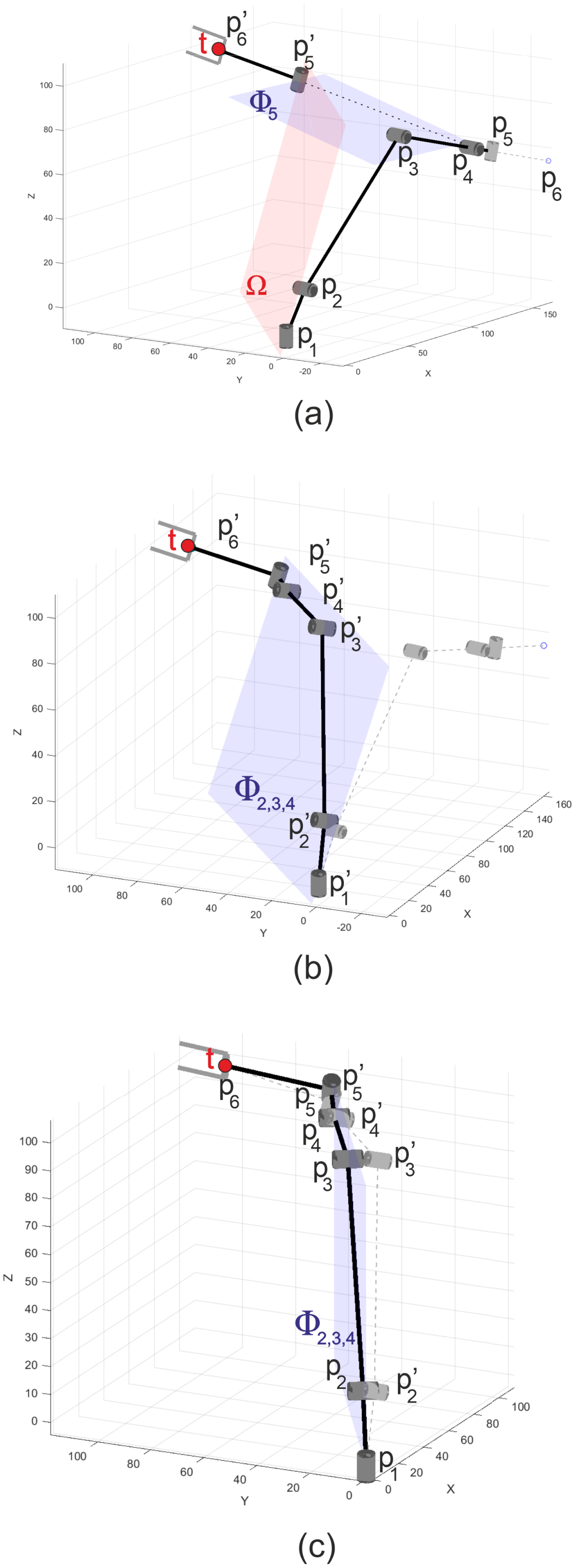

Fig. 5. Adaption of FABRIK to insert the joint movement constraints.

Fig. 6. Test setup for bottle picking experiment in dry lab with the underwater manipulator Schilling Titan 2.

TABLE I
DH PARAMETERS FOR SCHILLING TITAN 2

Joint	θ (rad)	d (m)	a (m)	α (º)
1	θ_1	0.195	0.121	-90
2	θ_2	0	0.851	0
3	θ_3	0	0.483	0
4	θ_4	0	0.133	90
5	θ_5	0	0	90
6	θ_6	0.377	0	0

to reach the target. The execution was measured in MATLAB running on an Intel Core i5 1.8 GHz processor with 6GB RAM. No optimizations were used for the methods reported in the table.

TABLE II
AVERAGE RESULTS (FOR 1000 RUNS) FOR A TOLERANCE OF 1 cm.

Performance metrics	FABRIK-R	Pseudo-inverse
Number of iterations	2.54	990.13
Matlab exec. time	0.010 s	0.234 s

TABLE III
AVERAGE RESULTS (FOR 1000 RUNS) FOR A TOLERANCE OF 1 mm.

Performance metrics	FABRIK-R	Pseudo-inverse
Number of iterations	4.75	4960.02
Matlab exec. time (ms)	0.014 s	1.149 s

Table II shows that the proposed approach is more than 20 times faster than the second order Pseudo-inverse used in [6], being able to solve the inverse kinematics problem with an average of only 10 ms. FABRIK-R also needs fewer iterations to reach the target, with an average value of only 2.54 iterations against 990 of the Pseudo-inverse approach. Therefore, using FABRIK-R gives much better performance for the PBVS application. We also performed the same experiment varying the error tolerance from 1 cm to 1 mm. The results presented in table III show that FABRIK-R is less susceptible to a higher degree of tolerance error, with a difference of only 4 ms from the time needed to find a solution for 1 cm to 1 mm of tolerance. In terms of number of iterations, the difference was only 2.2 iterations on average. On the other hand, the second order Pseudo-inverse needed 4 times more iterations and increased approximately 900 ms for the error tolerance varying from 1 cm to 1 mm. It is important to highlight that it takes more than 1 s for the Pseudo Inverse algorithm to find a solution with a tolerance of 1 mm due to the significantly higher number of iterations and execution time.

Since the approach based on FABRIK-R was proven to be a faster solution for the inverse kinematics problem, we performed another simulation experiment to verify the algorithm performance. The task was for the end-effector to follow a sinusoidal path composed of 10 points, as presented in Fig. 7. In this experiment, FABRIK-R needed only 27 iterations and 170 ms to complete the entire path.

To confirm the developed method, we applied it on the Titan 2 manipulator in dry conditions, at the Centre of Robotics and Intelligent Systems (CRIS) at the University of Limerick, Ireland.

The performed experiment consisted of a pick and place application with the manipulator starting in the pose presented in Fig. 6. The end-effector started at a position with the following coordinates: $x = 0.5$ m, $y = 0.03$ m, $z = -0.1$ m. The global coordinate frame was set to the base of the robot and the target position to a bottle placed at the point: $x = 1.5$ m, $y = -0.3$ m, $z = -0.05$ m. After picking the bottle, the robot move was to the point $x = 1.5$ m, $y = -0.6$ m, $z = -0.05$ m. Matlab was used to execute the developed algorithm by computing the joint angles to move the robot to the desired poses. As presented in Fig. 8 and Fig. 9, the robot successfully picked up and moved the bottle. Other different target points were tested and the algorithm converged for all tests performed.

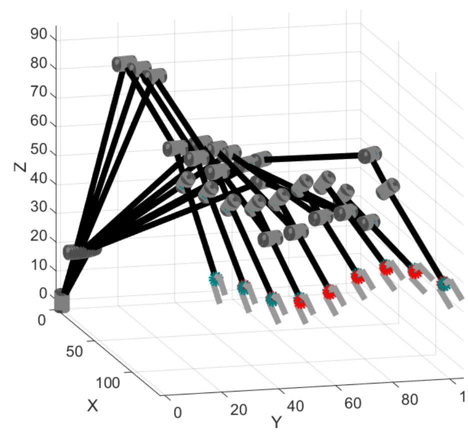

Fig. 7. Simulation of a path tracking task using FABRIK-R.

Fig. 8. Titan 2 reaching the first target point to pick the bottle.

Fig. 9. Titan 2 reaching the second target point to handle the bottle.

VI. Conclusion

This paper has presented the application of an extended version of the algorithm FABRIK for a highly constrained underwater manipulator Schilling Titan 2, comparing the performance with a classical numerical approach.

We proposed to solve the inverse kinematics based on the algorithm FABRIK-R, which uses the joint with a different actuation plane to generate the constraint planes for both Forward and Backward Reaching stages. This algorithm overcomes the problem of defining the constraint planes required in FABRIK when the chain is composed by joints of only 1DOF.

The main advantages of the proposed approach are the fast convergence and that it requires a low number of iterations to solve the inverse kinematics problem. This solution was compared with the Second order Pseudo-inverse Jacobian in simulation and results indicate advantages of the use of this FABRIK method for visual servoing control tasks.

The approach was also successfully applied in real experiments, in which the robot performed a pick-and-place task correctly based on the poses generated by our algorithm.

Future works may include experiments in underwater environments, including the use of visual feedback for target detection. Moving targets may also be considered, so the algorithm can be applied for tracking tasks.

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